

Gauss's 17-gon

Only a compass and a straightedge. Every point is marked on the front.

1. Quarter an angle

I is a quarter of the way from O to B. Bisect the angle at I between IO and IA, then bisect that half again. Where the line meets line OA: E.

2. 45 degrees, then a circle

At I, make a right angle to IE. Mark two points on line IE, one on each side of I, the same distance away. From each, draw a wider arc. Join I to where the arcs cross.

Bisect that right angle to get 45 degrees. Where it meets line OA, past O: F.

Find the middle of AF: equal arcs from A and from F, then join their crossings. The middle is M. Compass on M, opened to A: draw the circle. It meets OB at K.

3. One more circle

Compass on E, opened to K: draw the circle. It meets line OA at N3 and N5. Raise a perpendicular at each, up to the big circle: P3 and P5.

P3 is exactly $\frac{3}{17}$ of the way round from A. P5 is exactly $\frac{5}{17}$.

4. Walk it round

Open the compass from P3 to P5 and step it from A: you land $\frac{2}{17}$ of the way round. From there to P3 is one side, exactly $\frac{1}{17}$. Walk that width round the circle 17 times and join the marks. Every corner should land on a tick.

Gauss proved this possible in 1796, at 18. Richmond found this recipe in 1893.